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标题	PTML models of self assembled ligand free nanoparticle catalysts for cross coupling reactions
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摘要（原文节选）

Cross-coupling reactions have transformed the synthesis of complex and valuable compounds used in pharmaceuticals, materials science, and chemical synthesis. Transition metal nanoparticle (NP) catalysts represent a promising strategy within this field, but their behavior and efficiency continue under investigation. The use of computational models enables rapid design, optimization, and understanding of the behavior of these molecules, thereby reducing the costs and time. In this study, the perturbation theory and m ⋯

数据表 · Table 2（8 行，含数值行 8 行）

Training and test MAE and RMSE values of the MLP (9:9 – 20 – 9 – 1:1) and RBF (9:9 – 70 – 1:1) models for yields below and above 79%.

Model	Dataset	Yield range (%)	MAE (%)	RMSE (%)
MLP 9:9-20-9-1:1	Training set	> 79	3.77	5.39
≤ 79	12.39	17.91		
Test set	> 79	3.67	5.46	
≤ 79	13.63	17.97		
RBF 9:9-70-1:1	Training set	> 79	3.55	5.54
≤ 79	12.31	17.99		
Test set	> 79	3.60	5.74	
≤ 79	13.20	17.61		

正文对应句（逐字）：In addition, given this data distribution, which could cause bias in the analysis, additional evaluations were performed by calculating the MAE and RMSE values for yields below and above 79%—the average yield of the dataset —to provide a more comprehensive assessment of each performance metric.

数据表 · Table 3（34 行，含数值行 34 行）

Overview of ML models for predicting yields in cross-coupling reactions.

Model ’ s reactions	Catalyst metals	ML architectures	Params	Test values (%)	Featurization method	References
Suzuki – Miyaura, Kumada, Negishi, Buchwald-Hartwig, and C(sp2)- and C(sp3)-H functionalization	Pd, Fe, Ni, and Ru	PTML-MLR	RMSE	12.24	DRAGON molecular descriptors	This work
PTML-MLP	9.79					
Suzuki – Miyaura, Kumada, Negishi, Buchwald-Hartwig, C(sp2)- and C(sp3)-H functionalization, and double carbonylation	PTML-MLP	Accuracy	93.40			
Suzuki, Sonogashira, Cadiot-Chodkiewicz, Ullmann, and Buchwald-Hartwig	Mn, Fe, Co, Cu, Pd, and Zn	KNN (HTP)	Accuracy	60.00	RDKit FP	23
66.00	DRFP					
59.00	RXNFP					
NN	RMSE	14.23	DRFP			
15.18	RDKit FP					
20.50	RXNFP					
RF	13.56	DRFP				
14.33	RDKit FP					
19.95	RXNFP					
Suzuki, Sonogashira, Cadiot-Chodkiewicz, Ullmann, and Buchwald-Hartwig	Mn, Fe, Co, Cu, Pd, and Zn	MPNN	RMSE	14.55	RDKit molecular descriptors	24

Model ’ s reactions	Catalyst metals	ML architectures	Params	Test values (%)	Featurization method	References
ResGCN	14.62					
GATv2	15.00					
GraphSAGE	14.56					
GCN	16.05					
GAT	16.18					
GIN	24.28					
Suzuki	Pd	NN	Accuracy	74.30	OHE	26
76.40	RDKit FP					
73.90	DFT					
RF	73.90	OHE				
76.80	RDKit FP					
76.60	DFT					
xGB	73.70	OHE				

脚注（逐字）：PTML, Perturbation Theory Machine Learning; MLR, Multiple Linear Regression; LR, Linear Regression; MLP, Multi-Layer Perceptron; KNN, k-Nearest Neighbours; HTP, Hyperparameter Tuning; NN, Neural Network; RF, Random Forest; MPNN, Message Passing Neural Network; ResGCN, Residual Graph Convolutional Ne

正文对应句（逐字）：In addition, given this data distribution, which could cause bias in the analysis, additional evaluations were performed by calculating the MAE and RMSE values for yields below and above 79%—the average yield of the dataset —to provide a more comprehensive assessment of each performance metric.

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